

FL-01: AI Workflow Audit and Tool Setup

Student Profile: B.Tech AI & ML Engineering Student | Track: Applied Search Intelligence & ML

1. AI Workflow Audit Table (15 Tasks)

#	Task	Freq.	Classification	One-line Rationale
1	Solving LeetCode & DSA Problems	Daily	Just Me	Problem-solving intuition under timed conditions must be built unassisted for technical interviews.
2	Live Technical Coding Interviews	Weekly	Just Me	Real-time evaluations require unassisted personal reasoning, communication, and coding under pressure.
3	Reading ML Research Papers	Weekly	Collaborate with AI	AI accelerates comprehension of complex math/architectures while I synthesize core insights.
4	ML Model Development	Weekly	Collaborate with AI	AI assists in brainstorming features and model selection while I design leak-free validation.
5	Full-Stack Web App Development	Weekly	Collaborate with AI	AI pairs on UI components and API endpoints so I focus on system architecture and state.
6	Learning AI Frameworks/Libraries	Weekly	Collaborate with AI	AI provides interactive explanations and code examples for tools (PyTorch, DuckDB, LangChain).
7	Planning Capstone Project	Bi-weekly	Collaborate with AI	AI evaluates project feasibility and literature gaps while I define business objectives and scope.
8	Debugging Python & ML Code	Daily	Delegate to AI w/ Review	AI rapidly identifies syntax errors and shape mismatches, which I verify before applying to codebase.
9	Writing Technical Documentation	Weekly	Delegate to AI w/ Review	AI drafts API docs and setup guides from raw code, which I review for precision and accuracy.
10	Writing Research Notes	Weekly	Delegate to AI w/ Review	AI organizes experimental notes into Markdown summaries, which I verify against experiment logs.
11	Hackathon Project Scaffolding	Monthly	Delegate to AI w/ Review	AI generates initial project boilerplate and styling templates under tight deadlines subject to review.
12	Resume & Portfolio Optimization	Monthly	Delegate to AI w/ Review	AI tailors bullet points for technical impact, which I verify against actual project contributions.
13	LinkedIn Technical Posts	Bi-weekly	Delegate to AI w/ Review	AI outlines post drafts based on technical project milestones, which I refine for authenticity.
14	GitHub Project Management	Weekly	Fully Automate	Automated scripts generate release notes, format PR templates, and triage issue labels.
15	Internship Application Tracking	Weekly	Fully Automate	Automated scripts log submission dates, status updates, and metadata into a tracking dashboard.

2. Three Target Tasks for 8-Week AI Integration

Target Task	Why it Matters & Success Definition	Measurable Completion Criteria
1. Debugging Python & ML Pipeline Code	Debugging tensor shape mismatches, memory errors, and pandas bugs consumes 40% of dev time. Success means rapid root-cause diagnosis without swallowing exceptions.	Time Saved: MTTR reduced from 45 min to <10 min (75%+ reduction). Accuracy: 90%+ first-pass resolution rate without regressions. Production-Ready: Zero silent fallbacks or swallowed exceptions.
2. ML Model Dev & Feature Engineering	Building ML models requires fast iteration over features and architectures. Success means systematic evaluation while maintaining strict client-holdout validation discipline.	Fewer Revisions: 50% fewer iteration cycles via AI baseline auditing. Production-Ready: 100% adherence to leak-free pipelines. Understanding: Ability to defend model trade-offs in write-ups.
3. Reading & Summarizing ML Research Papers	Staying current with search intelligence and LLM literature requires parsing dense papers. Success means extracting core formulas and architectures in <15 mins per paper.	Time Saved: Paper analysis cut from 2 hrs to 25 mins (80% reduction). Understanding: High-fidelity synthesis for PyTorch implementations. Quality: Zero hallucinated claims or overstated findings.

3. Claude Project Custom Instructions

You are acting as an expert AI/ML engineering mentor, technical code reviewer, and tutor for a B.Tech CS student specializing in AI & ML. Key Guidelines: 1. Focus on ML, Deep Learning, Search Intelligence & Full-Stack (Python, PyTorch, pandas, scikit-learn). 2. Always explain first-principles intuition & trade-offs BEFORE code solutions. 3. Tone: Concise, technically rigorous, precise. 4. Code Standards: Clean, modular PEP8 Python with type hints & docstrings. Never swallow exceptions. 5. Analyze Big-O time/space complexity explicitly. 6. Enforce strict scientific discipline (correlation vs causality, leak-free validation).

4. Manual Steps Remaining Checklist

- ☐ Create Claude account (claude.ai) | ☐ Create Anthropic Academy account (academy.anthropic.com)
- ☐ Enroll in AI Fluency: Framework & Foundations | ☐ Complete Module 1 of AI Fluency framework
- ☐ Create Claude Project 'FlyRank ML Internship' | ☐ Paste custom instructions into Project
- ☐ Take screenshot of Claude Project interface | ☐ Take screenshot of Anthropic Academy enrollment

5. Assignment Rubric Self-Check

Criteria	Status	Verification Detail
At least 10 tasks in audit table	PASS	15 recurring tasks across academic, engineering, and career workflows.
Every task classified & rationalized	PASS	Uses 4 standard categories; 100% of tasks have concise rationales.
At least two 'Just Me' tasks	PASS	2 tasks designated Just Me (LeetCode/DSA & Live Technical Interviews).
Three target tasks defined	PASS	Detailed breakdown for Debugging, Model Dev, and Paper Reading.

Measurable completion criteria	PASS	Metrics specified (75% MTTR reduction, 50% fewer iterations, 80% time saved).
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